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Data Analytics, Data Science & Visualization with Python (48 hours 16 Session)

Python Basics

- What is Python
- Application of Python
- Why use Python for AIML Installation
- Anaconda/Other Idle
- Python Tokens
- Data Types in Python
- Conditional Statement
- Loops in Python
- Functions in Python
- Advance Functions
- File Handling
- **Mini Project**

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Intro to Git & Github and Kaggle

- Git and GitHub
- kaggle

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Python NumPy & Pandas

- Introduction to NumPy
- Exploring a NumPy Array
- Indexing & Slicing a NumPy Array
- Manipulating a NumPy Array
- Performing Mathematical & Statistical Functions using NumPy
- Performing Linear Algebra Operations using NumPy
- Introduction to Pandas
- Exploring Pandas Series
- Introduction to Pandas DataFrame
- Importing & Exporting Data
- Implementing basic DataFrame functionalities
- Exploring Descriptive Statistics with Pandas
- **Mini Project**

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R Language Essentials

- Fundamentals of R
- Vectors & control statements
- Functions in R
- Matrices & strings
- Lists & arrays in R
- Data visualization in R

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Exploratory Data Analysis (EDA)

- Data Types
- Dispersion & Skewness
- Data imputation
- Data Pre-processing
- Data Cleaning
- Data Manipulation
- Advanced Manipulation
- **Mini Project**

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SQL For Data Analytics

- SQL Basic
- SQL Joins
- SQL Aggregations
- Subqueries and Temp Tables
- SQL Data Cleaning
- Window Functions
- **Mini Project**

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Data Analysis with Excel

- Key Formulas and Functions, Ranges and Tables
- Data Cleaning – Text Functions, Dates and Times
- Conditional Formatting, Sorting and Filtering
- Dashboard Creation
- Analysis with Pivot Tables
- Data Analysis in Excel – Trends and Patterns
- Data Visualization in Excel – Charts and Plots
- Functions in Python
- Advance Functions File Handling
- **Mini Project**

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Data Visualization with Python

- Why Data Visualization?
- Introduction to Data Visualization
- Libraries & Tools for Data Visualization in Python
- Static Data Visualization Using Seaborn
- Interactive Data Visualization Using Plotly Express
- Interactive Animations & Facet Plots

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Tableau for Business Intelligence

- Understanding Data
- Creating Your First visualization
- Tableau Calculations
- Formatting Visualizations
- Manipulating Data in Tableau
- Creating Dashboards AND Stories
- Distributing & Publishing Your Visualization
- **Mini Project**

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Visual Storytelling using Power BI

- Introduction To Power BI
- Creating Power BI Reports, Auto Filters
- Report Visualization And Properties
- Chart And Map Report Properties
- Hierarchies And Drilldown Reports
- Power BI Deployment & Cloud
- Improving Power BI Reports And More...
- **Mini Project**

Capstone Project- 1 in Data Analytics